

AI-Driven Hospital Resilience

A Unified Machine Learning Framework for NHS Operational Optimization

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Executive Summary

This project addresses the critical operational pressures facing the NHS, characterized by chronic bed occupancy rates exceeding 95%. By leveraging clinical and operational data from the MIMIC-IV dataset (n=109,206), a unified machine learning pipeline was developed to predict three key outcomes: ICU deterioration, Length of Stay (LOS), and hospital readmission. The framework demonstrates that predictive intelligence can recover approximately **34,340 bed-days** annually, translating to a potential cost-avoidance of **£56.5 million**.

Technical Methodology

The system utilizes a 24-hour "Patient Velocity" window, capturing the rate of change in physiological vital signs and laboratory results. The unified MLOps pipeline employs diverse architectures tailored to specific outcomes:

- **ICU Escalation & Readmission:** XGBoost Classifiers optimized for clinical safety.
- **Length of Stay (LOS):** A Multi-Layer Perceptron (MLP) Regressor with a (128, 64, 32) hidden layer architecture to capture non-linear recovery patterns.

Key Metrics:

- ICU Deterioration: **AUROC = 0.9548**
- Length of Stay: **MAE = 1.41** days
- Hospital Readmission: **AUROC = 0.668** (Threshold: 0.30)

Economic & Operational Impact

The primary contribution of this research is the "Economic-Technical Bridge," converting abstract statistical performance into actionable hospital capacity metrics.

Operational Metric	Statistical Driver	Bed-Day Recovery	Fiscal Impact
ICU Escalation Prevention	0.9548 AUROC	21,840 Days	£43.68m
LOS Optimization	1.41-day MAE	8,750 Days	£4.375m
Readmission Reduction	0.668 AUROC	3,750 Days	£8.40m
Total Combined Impact	Unified Pipeline	34,340 Days	£56.45m

The recovery of 34,340 bed-days is equivalent to adding **94 virtual beds** or creating three new wards without infrastructure capital expenditure.

Transparency & Governance

To foster clinical trust, the framework integrates **SHAP (SHapley Additive exPlanations)** to provide patient-level interpretability. Key features driving predictions include medication counts, unique lab items, and urine output trends. The deployment blueprint emphasizes:

- **Drift Monitoring:** Continuous auditing for performance decay.
- **Bias Audits:** Ensuring algorithmic equity across patient demographics.
- **MDT Integration:** Supporting Multidisciplinary Team decisions as a clinical decision-support tool.